

week 1

Session 5

Task 1:

```
import java.util.Arrays;

class Solution {

    public int[] sortedSquares(int[] nums) {

        int n = nums.length;

        int[] result = new int[n]; // Initialize the result array of the same size

        int left = 0;           // Pointer for the start of the array

        int right = n - 1;      // Pointer for the end of the array

        int k = n - 1;         // Pointer for filling the result array from the end

        // Iterate while the left pointer is less than or equal to the right pointer

        while (left <= right) {

            // Compare the absolute values of the elements at the left and right pointers

            if (Math.abs(nums[left]) > Math.abs(nums[right])) {

                // If the absolute value of the left element is greater, square it and place it at the current end of the result
array
                result[k] = nums[left] * nums[left];

                left++; // Move the left pointer one step to the right

            } else {

                // Otherwise, square the right element and place it at the current end of the result array

                result[k] = nums[right] * nums[right];

                right--; // Move the right pointer one step to the left

            }

            k--; // Move the result array pointer one step to the left

        }

    }

}
```

```
        return result;
    }
}
```

Accepted Runtime: 0 ms

✔ Case 1

✔ Case 2

Input

```
nums =
[-4,-1,0,3,10]
```

Output

```
[0,1,9,16,100]
```

Expected

```
[0,1,9,16,100]
```

Task 2:

```
class Solution {

    public int pivotIndex(int[] nums) {

        int totalSum = 0;

        for (int num : nums) {

            totalSum += num;

        }

        int leftSum = 0;

        for (int i = 0; i < nums.length; i++) {

            if (leftSum == totalSum - leftSum - nums[i]) {

                return i;

            }

        }

    }

}
```

```
    leftSum += nums[i];  
  
}  
  
return -1;  
  
}  
  
}
```

Accepted Runtime: 0 ms

✓ Case 1 ✓ Case 2 ✓ Case 3

Input

```
nums =  
[1,7,3,6,5,6]
```

Output

3 ●

Expected

3